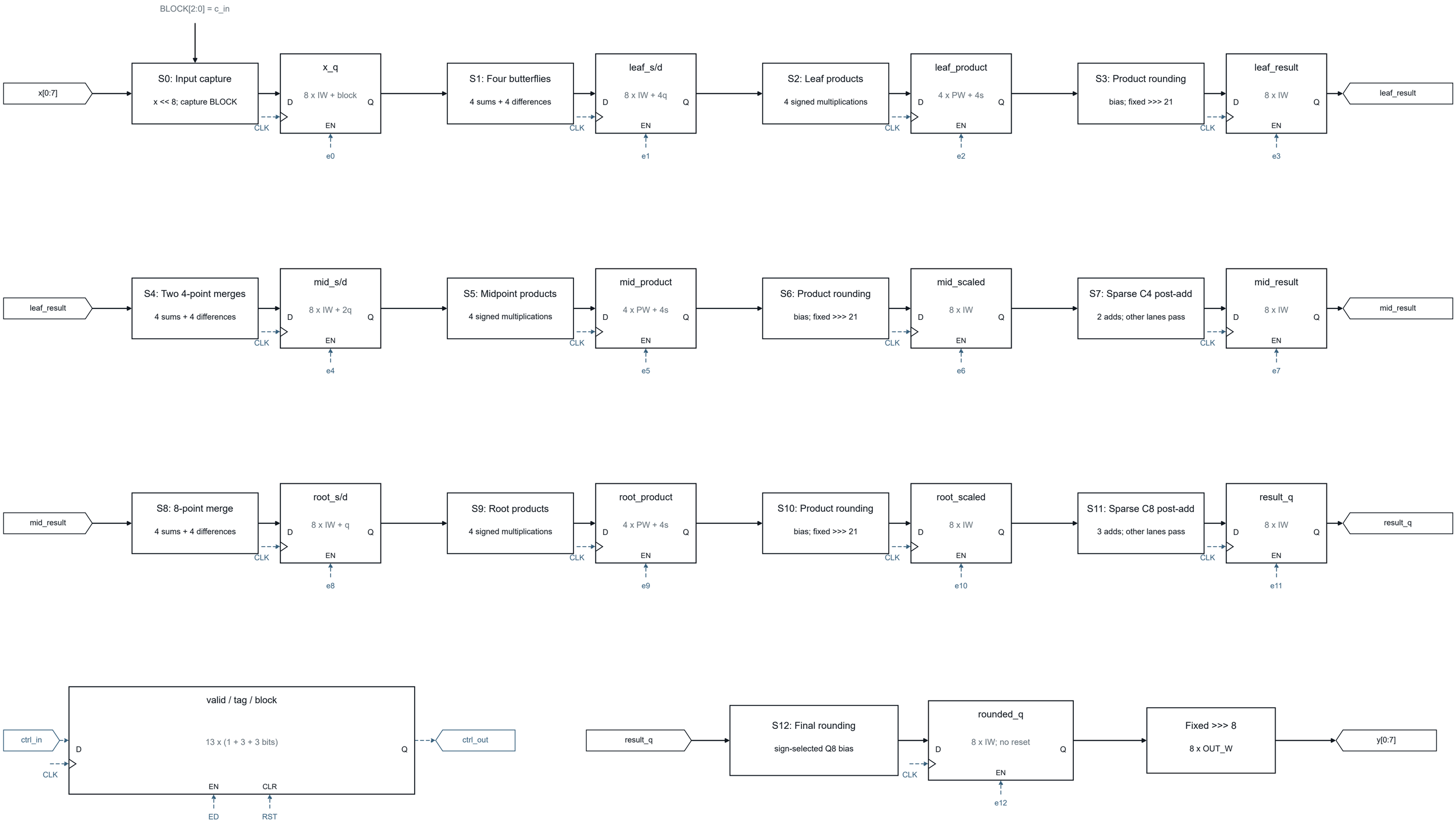
SkewDCT8 core - recursive arithmetic with registered boundaries

USE_COEFF_PORT=2: c_in is the 3-bit BLOCK ID. 12 real products; II=1 enabled edge; unchanged 13-stage arithmetic.



IW=DATA_W+16 (8 fractional guard bits); PW=IW+27; factors are signed Q21. Data registers have no reset; synchronous RST clears control state.

e0 = ~RST & ED & in_valid; ej = ~RST & ED & valid_q[j-1]. ctrl_in = in_valid, in_tag, BLOCK; ctrl_out = out_valid, out_tag.